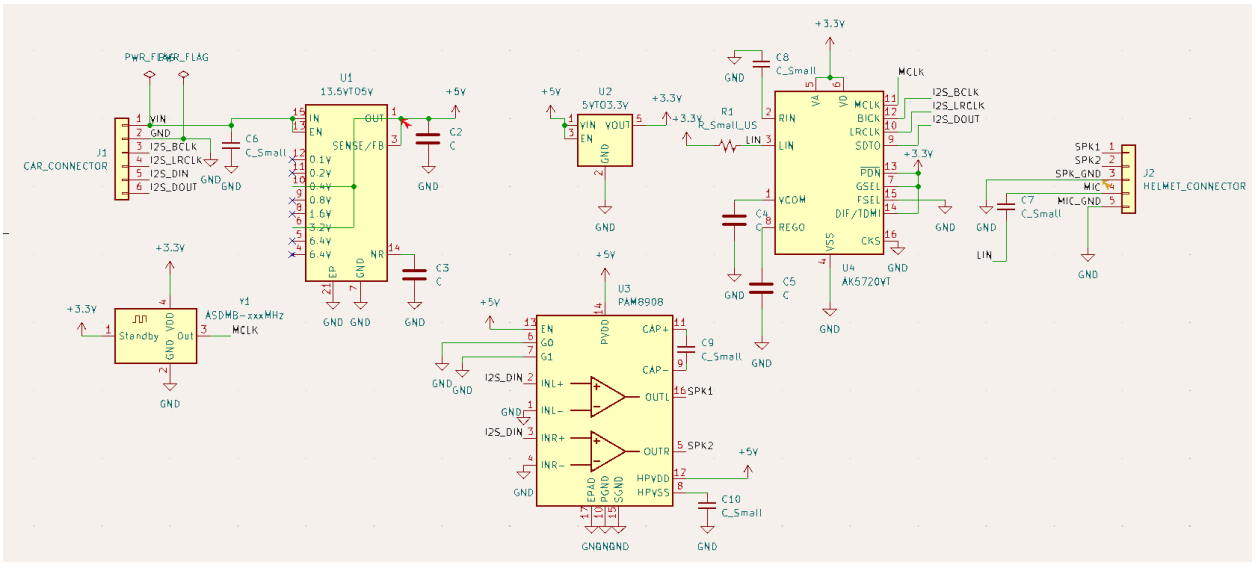


	Type	Manufacturer	Part Number	Digikey link	Digikey part number	Cost - 1 part	Frequency	Resonant Freq	Power	Impedance	Efficiency	Dimensions
2	speaker	Ole Wolff Electronics Inc	OWR-3507T-36A	https://www.digikey.com/en/produ	3949-OWR-3507T-36A-ND	$5.56	20 Hz ~ 20 kHz	72Hz	10 mW rated, 20 mW Max	36 ohm	97.4 dB @ 10mm & 10mW	Round, 35mm D x 7.82mm H
3	speaker	Ole Wolff Electronics Inc	OWR-5088TA-32	https://www.digikey.com/en/produ	3949-OWR-5088TA-32-ND	$4.15	20 Hz ~ 20 kHz		30 mW rated, 50 mW max	34 ohm	97 dB @ 10mm & 10mW	Round, 50mm D x 8.8mm H
4	speaker	PUI Audio, Inc.	AR03032MR-R	https://www.digikey.com/en/produ	AR03032MR-R-ND	$3.34	60 Hz ~ 10 kHz		20 mW rated, 30 mW max	32 ohm	115 dB @ 10mm & 1mW	Round, 29mm D x 7.3mm H
5	speaker	PUI Audio, Inc.	AS03808PR-R	https://www.digikey.com/en/produ	668-1419-ND	$3.87	800 Hz ~ 5 kHz		100 mW rated, 200 mW max	8 ohms	89 dB @ 100mm & 100mW	Round, 38mm D x 9.2mm H
6												
7	Type	Manufacturer	Part Number	Digikey link	Digikey part number	Cost - 1 part	Frequency	Current	Voltage	Impedance	Sensitivity	Dimensions
8	microphone	PUI Audio, Inc.	PNM-5050L-C3310-R	https://www.digikey.com/en/produ	668-1674-ND	$2.95		500 microA	2V rated		-50dB ±4dB	Round, 6mm x 5.2mm
9	microphone	Mallory Sonalert Products Inc.	PMOF-9745W-42DQ	https://www.digikey.com/en/produ	458-PMOF-9745W-42DQ-ND	$2.71	20 Hz ~ 16 kHz	500 microA	3V rated, range 1.1V ~ 10V	2.2 kohm	-42dB ±3dB @ 94dB SPL	Round, 9.7mm x 5mm
10	microphone	Same Sky (Formerly CUI Devices	CMEJ-4622-25-L082	https://www.digikey.com/en/produ	102-6539-ND	$2.48	100 Hz ~ 10 kHz	500 microA	3V rated		-25dB ±3dB @ 94dB SPL	Round, 4.6mm x 2.4mm
11	microphone	Knowles	FB-EM-30342-000	https://www.digikey.com/en/produ	FBEM30342000-ND	$68.29	300 Hz ~ 6 kHz	50 microA	1.3V rated, range 1.3V ~ 10V	4.4 kohm	-50dB ±4dB @ 74dB SPL	Round, 19 mm x 6.22 mm
12												
13	Type	Manufacturer	Part Number	Digikey link	Digikey part number	Cost - 1 part	Contacts	Pitch	Connection			
14	connector	Adafruit Industries LLC	5359	https://www.digikey.com/en/produ	1528-5359-ND	$8.95	5	2.54mm	Magnetic snap-connector			
15	connector	Adafruit Industries LLC	5413	https://www.digikey.com/en/produ	1528-5413-ND	$8.95	5	2.54mm	Magnetic snap-connector			
16												
17												
18	Type	Manufacturer										
19	I2C Codec	Texas Instruments	TLV320AIC3100IRHBR	https://www.digikey.com/en/produ	296-25494-1-ND	$3.73						
20												

Maheer Grang LV Driver Audio Project Writeup

This project basically aimed to design a compact audio interface for driver communications in our helmet. The goal is to take a digital I2S stream from the telemetry/radio bus and drive the helmet speakers while capturing clean microphone audio from the driver, while running off of the cars 13.5 volts

Component Selection & My Thinking

I looked at DigiKey to find parts that fit comfortably inside a helmet and survive a loud racecar environment. Here's what I chose and why:

1. **Speakers**: Ole Wolff OWR-3507T-36A
   Why: Most headphone drivers are massive (50+mm) and super thick. These are 35mm across and only 7.82mm thin, so they slide right into standard motorcycle helmet ear pockets without squishing the driver's ears.
   Acoustics: 38 ohm impedance with a clean 20hz - 20khz range focused on vocal clarity over heavy bass, which helps the communications beat the road noise.

2. **Microphone**: PUI Audio PNM-5050L-C3310-R
   Why: Tiny electret capsule (6mmx5.2mm) that fits easily inside the chin bar. It draws barely any power 500 microA and works nicely with a simple resistor bias and AC-coupling cap.

3. **Emergency Breakaway Connector:** Adafruit 5-Pin Magnetic Snap (5359 / 5413)
   Safety: FSAE rules require the driver to exit the car super fast in an emergency. Threaded or latched connectors are a hazard because you can get caught on the wire.
   Solution: A 5-pin magnetic pogo-pin connector that holds firm during driving but pops straight off if the driver jumps out.

Pinout Definitions

Radio / Car Side (J1)

- VIN: +13.5V Vehicle Bus Input
- GND: System Ground
- I2S_BCLK: Bit Clock Line
- I2S_LRCLK: Frame Clock / Word Select Line
- I2S_DIN: Incoming Speaker Audio Stream
- I2S_DOUT: Outgoing Mic Audio Stream

Helmet Side (J2 - Magnetic Snap)

- SPK1: Left Speaker
- SPK2: Right Speaker
- SPK_GND: Speaker Ground
- MIC: Helmet Microphone Input
- MIC_GND: Mic Ground

Why no microcontroller?

Luca said no :(

Schematic Component Rationale

Here is the breakdown of why each part on the board is there and how I designed the schematic:

1. Primary Regulator: TPS7A4701 LDO (U1)
   - Why: Buck converters create a lot of high-frequency electrical noise that bleeds straight into analog audio lines which messes with the audio quality. I picked the TPS7A4701 because it's a low-noise linear regulator made specifically for clean audio stages.

2. Secondary Regulator: AP2112K-3.3 LDO (U2)

- Why: This thing converts 5V down to 3.3V for the oscillator and the ADC. Running this on its own LDO keeps the digital noise separated from the main 5V rail I believe.

3. Active Clock Oscillator: Abracon ASDMB 12.288 MHz (Y1)
    - Why: We need a super accurate, jitter-free master clock to sample audio cleanly at 48 kHz (256x sampling rate) so I picked having an oscillator on the board

4. Audio ADC: AK5720VT 24-Bit ADC (U4)
    - Why: Digitizes the driver's voice from the helmet mic into I2S_DOUT. The best part about this chip is that it runs 100% on hardware, so it boots instantly without needing I2C as Luca told me to figure out.

5. Amplifier Stage Placeholder: PAM8908 (U3)
    - Why: Used as a stand-in to map out the physical layout for the speaker driver stage. I originally was going to use the MAX98357A speaker driver, but I couldn't find an existing KiCad footprint for it so this was the closest thing.

I still need to calculate the exact values for all the capacitors and resistors and figure out if im missing any, but I think this layout works for what we're trying to do right now and I have the basics down.Thank you for reading this!